

How does Global MPPT scan work?



In some cases, your photovoltaic arrays may be installed under shadow from surrounding buildings or plants, and the maximum power point(**MPP**) may get affected by such partial shading conditions for some hours during the day, resulting in decrease of energy production.

In order to solve this issue and increase the yields, SolaX has integrated Global MPPT(**GMPPT**) scan solution into the latest solar inverters.



How does Global MPPT scan work?

In normal cases, there is no shading on the photovoltaic arrays, thus SolaX inverters can reach the Maximum Power Point effortlessly and precisely by scanning from the right side(U_{oc} , Open Circuit Voltage) to the left till the Input Power **P** would start decreasing if it keeps moving left.

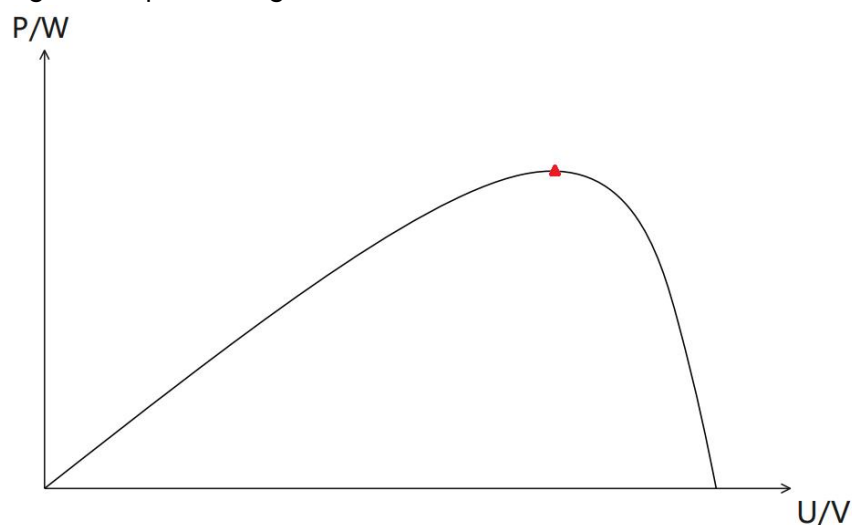


Fig. 1

However, when part of photovoltaic arrays is shaded, the MPP scan curve would become more complex with multiple maxima(Fig. 2). Since the MPP scan would stop when the Input Power **P** starts decreasing for the first time, the relative Maximum Power Point **P1** may not be the actual MPP on the whole curve. The scan logic would cause power reduction with more and more yield losses for the owner.

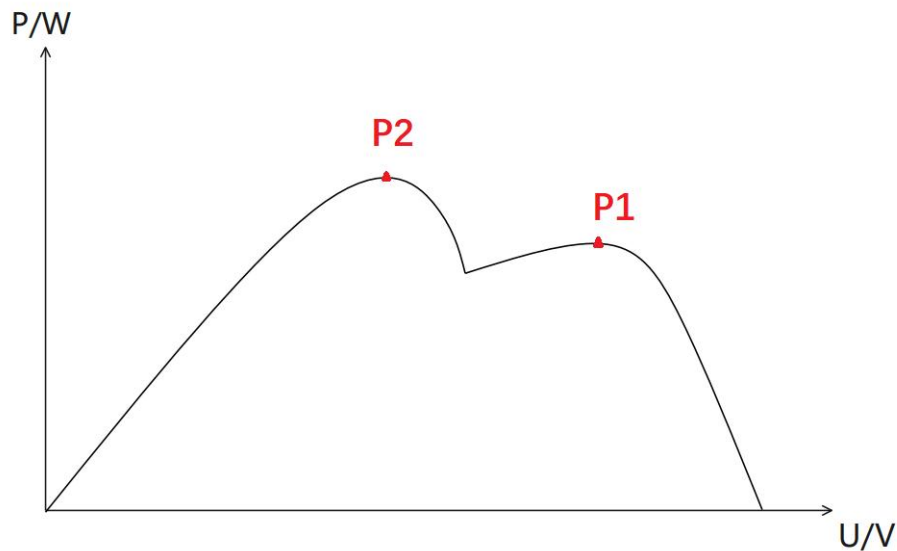


Fig. 2

On the other hand, with SolaX Global MPPT Scan solution, SolaX inverters will maintain an active MPP curve scan on a regular time basis, e.g. once in 1h, 3h or 4h. In each scan, a wide MPP range will be checked to make sure the GMPPT gets the actual highest Maximum Power Point possible.

After enabling **GMPPT** function and selecting scan interval for each PV input, it will check each power point from U_{oc} (Refer to **P1** on Fig. 3) to the left side(Fig. 3). Once the full range is checked, SolaX inverter will know multiple maxima on this curve, and quickly operate at the optimal voltage under **Pmax**(Refer to **P4** on Fig. 3)

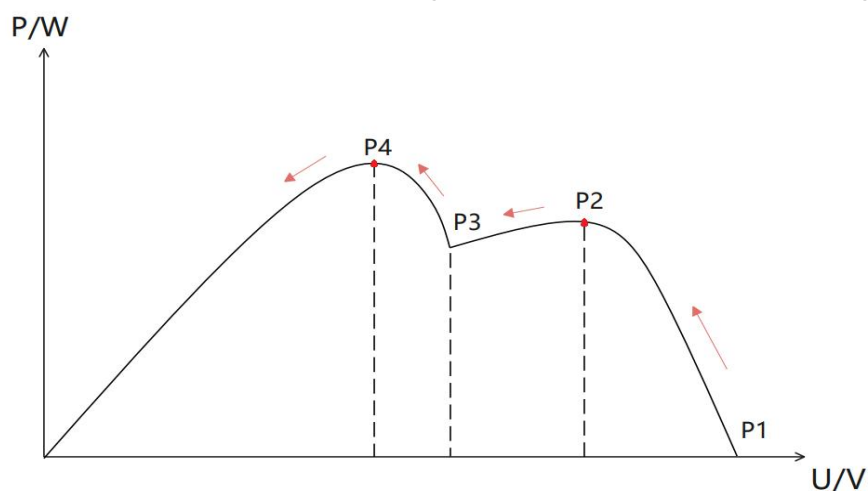


Fig. 3

Note: During GMPPT scan, the actual power would change with the scanning process.

How to Enable GMPPT function?

The following options shall be set in the inverter settings. The menu structure is as below:

Main Page → Setting → Enter Password → GMPPT → PV* Control

All independent MPP Trackers can be controlled separately in accordance to the actual environment. There are four options available for each PV Control:

PV* Control Options	Function	Notes
Off	Disable GMPPT	GMPPT scan will not be activated
Low	Set scan interval to 4h	GMPPT scan once every 4h
Middle	Set scan interval to 3h	GMPPT scan once every 3h
High	Set scan interval to 1h	GMPPT scan once every 1h

It's recommended to set via SolaX Cloud APP or Web like the following page

GMPPT

PV1 Control

Off Save

PV2 Control

Off Save

Cancel OK

Off

Low

Middle

High

Notes

1. For multiple MPP trackers, the GMPPT scan will be done one by one automatically.
2. GMPPT will not work when PV voltage falls lower than Min MPPT Voltage.
3. In case of an error or inverter restart, the GMPPT scan timer will restart once it's back to normal status.
4. If photovoltaic arrays are oversized, the GMPPT scan will stop at the limited power point.
5. GMPPT doesn't help optimize transient shading like slow-moving clouds.
6. GMPPT is set as **Off** by default.

Features

1. Independent GMPPT scan on each MPPT
2. Controllable GMPPT scan interval
3. Fast scan speed(<1min for single PV GMPPT scan, vary on different models and running conditions)
4. Solve fixed shading issue in actual installation environment
5. Remote management via SolaX Cloud

On which SolaX inverters GMPPT scan is available?

Single Phase/Three Phase	Series	Power Range
Single Phase	X1-MINI G4/X1-BOOST G4	0.6 ~ 6kW
	X1-MINI G4 Plus	5~6kW
	X1-SMART G2	5~10kW
	X1-HYBRID G4	3.7~7.5kW
	X1-IES	3~8kW
	X1-AC	3~5kW
	X1-Vast	5~10kW
	X1-SPT	3~7kW
	A1-HYBRID G2	3.8kw~7.6kW
	A1-AC G2	3.8kW~7.6kW
	JIESS-HB58/58-1 JIESS-HB58X/115/173	5.9kW

Three Phase	X3-MIC G2/X3-PRO G2 X3-MIC G2 LV/X3-PRO G2 LV	3 ~ 30kW
	X3-MGA G2/X3-FTH X3-MGA G2 LV/X3-FTH LV	20 ~ 150kW
	X3-FTH Plus	120~150kW
	X3-IES	5~15kW
	X3-Fit G4	6~15kW
	X3-Grand	300~350kW
	X3-Ultra	15~30kW
	J3-Ultra	30kW
	J3-Ultra LV	16.5kW
	X3-HYBRID G4 Pro	4~15kW
	X3-HYBRID G4	5 ~ 15kW